A Quantum Multimodal Neural Network Model for Sentiment Analysis on Quantum Circuits

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***Abstract*—This article proposes a quantum multimodal neural network (QMNN) model that can be implemented on parameterized quantum circuits (PQCs), providing a novel avenue for processing multimodal data and performing advanced multimodal sentiment analysis tasks. The comprehensive QMNN model is structured into four fundamental blocks: multimodal data preprocessing, unimodal feature extraction, multimodal feature fusion, and optimization. Through these blocks, multimodal data are initially preprocessed and encoded into quantum states. Subsequently, visual and textual features are extracted from the quantum states and are then integrated to learn the interactions between different modalities. Finally, the model parameters are fine-tuned to optimize the sentiment analysis performance. Simulation results confirm that QMNN surpasses state-of-the-art baselines, using significantly lower input dimensions and substantially fewer parameters than classical models. Furthermore, the entanglement, integrity, robustness, and scalability of the model are analyzed in depth. Internally, the strong entanglement within the multimodal fusion block enhances interactions between textual and visual features, and the integrity of the model reflects the indispensable contribution of each component to the overall performance. Externally, robustness ensures the model operates stably under noisy conditions and incomplete inputs, and scalability enables it to efficiently adapt to varying architectural depths and widths. The above simulation results and performance analyses showcase the comprehensive strength of our proposed model.**

***Impact Statement*—In recent years, quantum machine learning (QML) technologies, especially quantum neural networks (QNNs) that are implementable on parameterized quantum circuits (PQCs), have garnered widespread attention. QNNs have achieved remarkable results in processing single-modal data such as images, text, and videos. However, their potential in handling multimodal data and learning correlations between different modalities has not been thoroughly explored. This article proposes a quantum multimodal neural network (QMNN) model towards implementation on PQCs for performing multimodal sentiment analysis tasks. Despite the very limited number of stable qubits, basic linear quantum gates and preliminary operations provided by the currently available PQCs, the proposed approach is still useful and can effectively process large-scale, real-world multimodal data with the limited computing resources. This not only broadens the application range of QNNs but also significantly contributes to the advancement of QML technologies.**

***Index Terms*—Multimodal sentiment analysis, quantum multimodal neural network (QMNN), quantum neural network (QNN).**

I. Introduction

RECENT advances in quantum technologies have enabled the availability of noisy intermediate-scale quantum (NISQ) processors with stable computational capabilities [1], [2], [3]. This breakthrough has propelled quantum machine learning (QML) to a pivotal research topic in both academy and industry [4], [5], [6], [7], [8]. QML enjoys both the efficiency of machine learning techniques and the computational power of quantum systems, showing great potential to advance artificial intelligence [9], [10], [11]. As the essential model of QML, the quantum neural network (QNN) has been extensively investigated and has found successful applications across diverse domains [12], [13], [14]. These include quantum convolutional neural networks (QCNNs) for computer vision tasks [15], [16], quantum language models for natural language processing (NLP) [17], [18], and quantum graph neural networks for graph-structured data related tasks [19], [20], among others. For an in-depth exploration of QNNs and their applications, one can refer to comprehensive survey papers [21], [22], [23] and the references therein.

The rapid development of information technology has led to an explosive growth in data volume and a significant diversification in data types. This diversification is primarily manifested by the emergence of a large amount of multimodal data, encompassing various forms such as images, text, audio, and video [24], [25]. As a result, the processing and analysis of such data is no longer limited to classical unimodal methods. Instead, a deeper consideration of the interactions and connections between the different modalities is more often required. However, managing complex relationships inherent in the vast

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scale and diversity of multimodal data poses significant challenges for classical machine learning algorithms. As indicated in the previous discussion, QNNs have already made considerable advances in critical domains such as image and text processing [13], [26], due to the powerful data processing capabilities of quantum computing. In particular, the QNNs based on parameterized quantum circuits (PQC) [27] feature a layered architecture that not only facilitates the intuitive applications of quantum computing principles in machine learning but also enhances the interpretability of the networks. Moreover, the ease of parameter tuning substantially improves the efficiency of QNNs for real applications. In light of these advantages, it is naturally proposed that QNNs be employed to manage more complex multimodal data. To the best of our knowledge, existing research on QNNs has primarily focused on unimodal data processing [15], [16], [17], [18], [19], [20], with few studies exploring their capabilities in processing multimodal data. This observed gap has inspired our research into quantum multimodal neural networks (QMNNs) and their applications in multimodal sentiment analysis tasks.

The main challenges along this research are twofold: 1) to effectively extract features from text and images, it is necessary to implement arbitrary quantum unitary transformations on PQCs. Furthermore, to deeply explore the intrinsic relationships of multimodal features, forming strongly entangled quantum states on PQCs is crucial. However, the current availability of PQC platforms, which provides only a limited number of qubits, very basic linear quantum gates, and preliminary operations, presents significant difficulties in achieving these objectives; 2) although QNNs demonstrate inherent strengths in processing quantum data, their research in handling large-scale real-world data is still in its early stages. Real-world data often involves more complex characteristics, including higher dimensionality and intricate correlations. This complexity poses a substantial obstacle to effectively implementing QMNN for multimodal sentiment analysis tasks. In this article, we propose a QMNN model that is implementable on PQCs for multimodal sentiment analysis tasks. The model is structured into four main blocks: multimodal data preprocessing, unimodal feature extraction, multimodal feature fusion, and optimization. Specifically, in the multimodal data preprocessing block, images and texts from multimodal data are individually transformed into vector representations and subsequently encoded into quantum states. Then, the unimodal feature extraction block employs two quantum convolutional layer architectures to extract visual and textual features, respectively. The extracted features are integrated into the multimodal feature fusion block. Next, the fused features are read out, and the QMNN model is optimized. Finally, simulation results using real-world multimodal sentiment analysis datasets demonstrate that the proposed QMNN can efficiently perform multimodal sentiment classification tasks.

The main contributions of this research are summarized as follows:

1) A novel QMNN model implementable on PQCs is proposed for multimodal sentiment analysis, which achieves superior sentiment analysis performance by using only a limited number of qubits and shallow circuit depths.
2) In the proposed QMNN, unimodal feature extraction blocks with high efficiency are introduced to independently extract rich visual and textual features. A multimodal feature fusion block with strong entanglement is developed to integrate these features, enhancing the interactions between images and text.
3) Experimental results on public datasets demonstrate that the proposed model outperforms state-of-the-art methods, showcasing its entanglement and integrity through multimodal fusion exploration and ablation studies, as well as its robustness and scalability through quantum noise analysis and evaluation of its adaptability to varying architectural depths and widths.

The remainder of the article is organized as follows. Section II introduces the preliminaries and related research. Section III describes the QMNN model design in detail. Section IV is devoted to the presentation of simulation results and discussions. Finally, the article is concluded in Section V.

II. Preliminaries and Related Work

This section provides an overview of the fundamental concepts in quantum computing and QNNs, followed by a review of the related works in sentiment analysis.

A. Quantum Computing

1) Quantum Bit: A quantum bit, or qubit, is the fundamental unit of information storage and processing in quantum computing, serving a role analogous to that of the classical bit in classical computing. It is implemented physically through a two-level quantum system, and represented mathematically by a two-dimensional complex vector within the Hilbert space $\mathcal{H}$. Specifically, a single-qubit state can be expressed as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \tag{1}$$

where $|0\rangle$ and $|1\rangle$ are two basis states of a qubit. The coefficients α and β are complex numbers that satisfy the normalization condition $|\alpha|^2 + |\beta|^2 = 1$. Several qubits can be integrated into a multiqubit system, whose quantum state space is a tensor product of the quantum state of each qubit, which can be expressed as follows:

$$|\Psi\rangle = |\psi_0\rangle \otimes \cdots \otimes |\psi_{n-1}\rangle = \sum_{i=0}^{2^n-1} \alpha_i |i\rangle \tag{2}$$

where each $|i\rangle$ is a basis state of the n qubits, corresponding to the binary representation of the integer i.

2) Quantum Gate: Quantum computers are constructed utilizing quantum circuits that consist of wires and quantum gates to facilitate the transmission and manipulation of quantum information. The quantum gates manipulate the states of the qubits, which dictates the evolution of the overall quantum system. In particular, the evolution of an isolated quantum system can be described by a unitary transformation, represented by a unitary operator U that satisfies $UU^\dagger = I$, with $U^\dagger$ denoting the

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TABLE I
BASIC QUANTUM GATES AND THEIR REPRESENTATIONS

Quantum gate	Matrix representation	Graphical symbol
I	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	I
X	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	X
Y	$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$	Y
Z	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	Z
$R_x(\theta)$	$\begin{bmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix}$	$R_x(\theta)$
$R_y(\theta)$	$\begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix}$	$R_y(\theta)$
$R_z(\theta)$	$\begin{bmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{bmatrix}$	$R_z(\theta)$
CNOT	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	

conjugate transpose of U. When a quantum gate U is applied to an initial quantum state $|\Psi\rangle$, it results in

$$U|\Psi\rangle = U\sum_{i=0}^{2^n-1}\alpha_i|i\rangle = \sum_{i=0}^{2^n-1}\beta_i|i\rangle. \tag{3}$$

In quantum circuits, commonly utilized quantum gates include the single-qubit rotation gates R_x, R_y, R_z, and the two-qubit CNOT gate. Specifically, the rotation gates can be expressed as $R_x(\theta) = e^{-i\theta X/2}$, $R_y(\theta) = e^{-i\theta Y/2}$, and $R_z(\theta) = e^{-i\theta Z/2}$, where X, Y, and Z represent the corresponding Pauli gates. The matrix representations and graphical symbols of these gates are summarized in Table I.

3) Quantum Measurement: The quantum measurement can extract information from the quantum state. The most commonly used method in current PQCs is the projective measurement. Let O be a quantum observable, which can be represented by a Hermitian operator. According to the spectral decomposition theorem, O can be expressed as

$$O = \sum_k \lambda_k P_k \tag{4}$$

where λ_k denotes the possible outcome of each projective measurement on O, and P_k represents the projector onto the eigenspace of O corresponding to the eigenvalue λ_k. Suppose the quantum system is placed at the state $|\Psi\rangle$, then the expectation value of measuring the observable O is calculated as $\langle O\rangle = \langle\Psi|O|\Psi\rangle$.

B. Quantum Neural Network

QNNs have emerged as a central application of quantum computing, aiming to leverage the quantum computational advantage to address real-world problems. Here, we only discuss

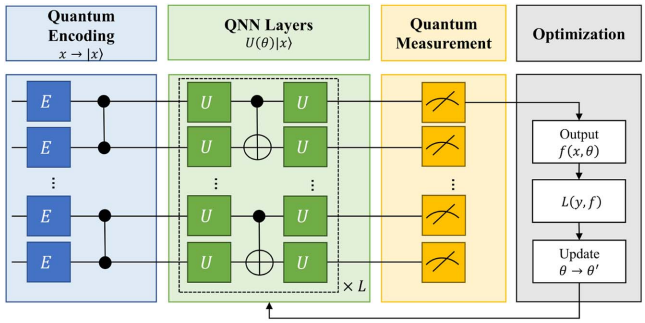


Fig. 1. Quantum neural network.

QNNs that are implementable on PQCs, as illustrated in Fig. 1. A QNN comprises four blocks: quantum encoding, QNN layers, quantum measurement, and optimization. First, the input classical information x is encoded as the quantum state $|x\rangle$ by the quantum encoding block. Then, the prepared quantum state is transformed via the QNN layers block $U(\theta)$ that consists of trainable unitary operators. Notably, each layer within this block is structured identically, satisfying $U(\theta) = \prod_{l=1}^{L} U_l(\theta_l)$ where L denotes the number of layers, and θ_l symbolizes the set of trainable parameters at the l-th layer. After executing $U(\theta)$, the quantum measurement block extracts the quantum information into the classical form, subsequently mapping the results to produce the output $f(x,\theta)$. Finally, the parameters θ are updated by minimizing the loss function $L(y,f)$, facilitating an iterative refinement of $U(\theta)$.

C. Related Works of Sentiment Analysis

Currently, few QNNs have been studied for sentiment analysis. Even though some quantum-inspired types of neural networks exist [28], [29], they differ significantly from our approach as they cannot be directly implemented on NISQ processors. Therefore, this subsection primarily discusses the classical related works that are directly relevant or have provided inspiration for our research.

1) Unimodal Sentiment Analysis:

a) Visual Sentiment Analysis: Early work in visual sentiment analysis primarily relied on handcrafted features such as texture and color, which was limited in capturing the complexities of visual emotions [30], [31], [32], [33]. More recently, particularly with convolutional neural networks (CNNs), deep learning models have marked significant progress in the field. In [34], a VGG model, initially trained on ImageNet and subsequently fine-tuned on sentiment-specific datasets, was employed to extract high-level image features, demonstrating the effectiveness of CNNs in visual sentiment analysis. In [35], DeepSentiBank, a visual sentiment concept classification model leveraging deep CNNs, was developed. Specifically, this model was initialized with weights trained on ImageNet and fine-tuned on a sentiment-tagged Flickr image dataset, which significantly improved classification accuracy. In [36], WSCNet, a weakly supervised coupled convolutional network, was introduced. This network focused on detecting sentiment maps

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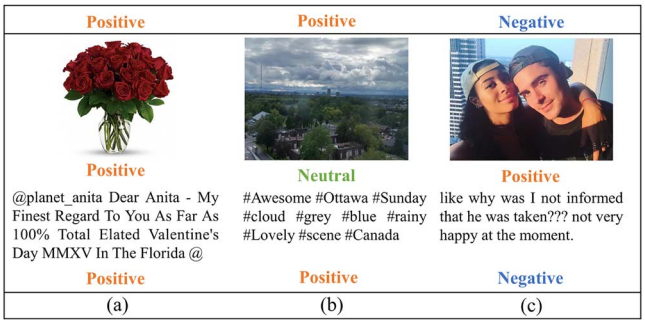


Fig. 2. Multimodal social media examples.

to capture local and holistic image features, enhancing visual sentiment analysis.

b) Text Sentiment Analysis: Initial research in text sentiment analysis focused on lexicon-based techniques and machine learning methods to identify sentiment polarity. These approaches were advantageous in capturing semantic features at low or middle levels [37], [38]. Inspired by the excellent results of deep learning models in NLP tasks, recent studies have increasingly employed advanced deep neural networks to capture sophisticated semantic features for sentiment analysis. In [39], a CNN for text sentiment analysis was first introduced, which used pretrained word vectors and minimal hyperparameter tuning to achieve outstanding text classification performance. In contrast to the single-layer CNN used for sentence-level tasks in [39], [40] demonstrated that deeper CNN architectures with multiple convolutional layers significantly enhanced document classification efficacy. Furthermore, recurrent neural networks have also yielded impressive results in sentiment analysis tasks, as detailed in several studies [41], [42], [43].

While unimodal sentiment analysis has significantly advanced, social media expressions often integrate textual and visual elements to convey sentiment, as shown in Fig. 2, presenting multifaceted challenges. In scenarios where text and image sentiments are congruent [as depicted in Fig. 2(a)], the sentiment expression can be reinforced. Conversely, when one modality remains neutral, and the other exhibits a clear sentiment, as shown in Fig. 2(b), the sentiment of the composite expression is typically guided by the polarized element. Furthermore, instances of polarity discordance between text and image [as demonstrated in Fig. 2(c)] introduce a more complex sentiment analysis. Therefore, the inherent multimodality of social media expressions necessitates more comprehensive approaches for accurate sentiment analysis.

2) Multimodal Sentiment Analysis: Based on the above discussion of unimodal sentiment analysis, this subsection focuses on presenting deep learning methods for multimodal sentiment analysis. In [44], a multimodal sentiment analysis framework was developed that leveraged deep CNNs to extract contextually rich features from textual data and used a facial recognition library to capture visual features from facial expressions in video frames. These features were combined through feature-level fusion, and a multiple kernel learning classifier was employed to handle the heterogeneous multimodal data effectively. In [45], a CNN architecture for multimedia sentiment analysis was proposed that extracted textual and visual features using pretrained text and image CNNs, respectively. These features were then combined by an additional CNN to obtain the intrinsic connection between the image and text, leading to improved sentiment prediction. In [46], a CNN-based multimodal approach was introduced that embedded texts into images to obtain information-enriched images, achieving superior accuracy over single modality and conventional multimodal strategies. Beyond CNN architectures, various deep learning models have been utilized for multimedia sentiment analysis. These include the cross-modality consistent regression method [47], long short-term memory networks [48], MultiSentiNet [49], and the image-text interaction network [25], among others. For more detailed discussions of multimodal models, readers are directed to survey papers [50], [51] and the references cited therein.

III. Quantum Multimodal Neural Network

This section delineates the architecture of the proposed QMNN for multimodal sentiment analysis. Initially, the problem is formalized, providing a foundation for the subsequent discussion. Then, preprocessing methods applicable to multimodal data are introduced. Subsequently, the QMNN framework is elucidated. Finally, the optimization of the QMNN is explicated.

A. Problem Formalization

In this article, the primary research goal is to develop a QMNN based on PQCs that excels in multimodal sentiment analysis. The system architecture, shown in Fig. 3, consists of essential blocks connected by a workflow indicated with purple arrows. Following this flow, we detail the algorithms implemented within each block.

1) Step 1: Input Data: The input data is a multimodal dataset represented as a collection of triples $\{(I_i, T_i, y_i)\}_{i=0}^{N-1}$, where I_i denotes an image, T_i represents the corresponding text, y_i is the sentiment label, and N indicates the number of image-text pairs.

2) Step 2: Data Preprocessing: Each image I_i and text T_i is processed into vector representations X_i^I and X_i^T through their respective preprocessing methods.

3) Step 3: Quantum Encoding: The preprocessed vectors X_i^I and X_i^T are encoded into a quantum state, represented as $|X_i\rangle$.

4) Step 4: Unimodal Feature Extraction: The encoded quantum state $|X_i\rangle$ is processed through unimodal feature extraction blocks, represented by the unitary operator $U(\theta_u)$, where θ denotes the tunable parameters. These blocks extract features from the input image-text pairs, and the process is mathematically expressed as $U(\theta_u)|X_i\rangle$.

5) Step 5: Multimodal Feature Fusion: The extracted features from the unimodal blocks are passed into the multimodal feature fusion block, represented by $U(\theta_M)$. This block integrates the image and text features to capture interactions

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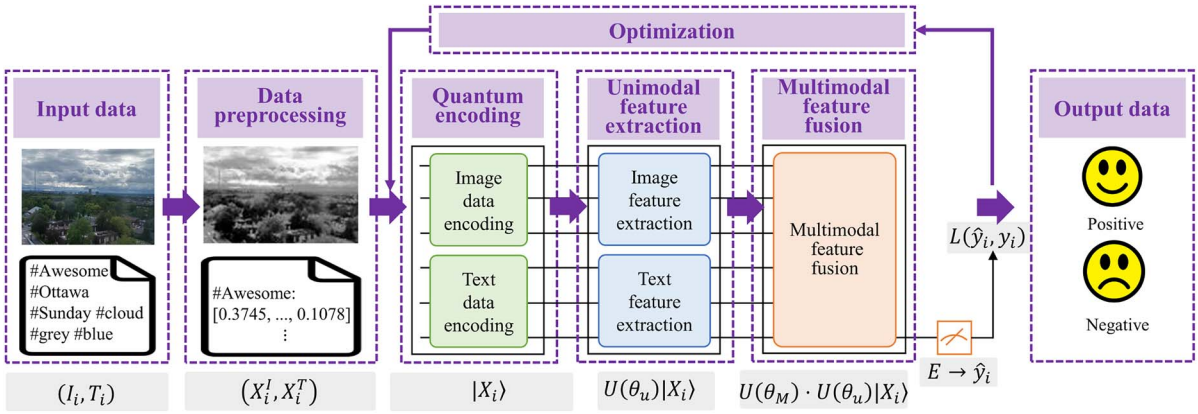


Fig. 3. System architecture of QMNN. All essential modules are highlighted in purple blocks, with the sequential workflow represented by purple arrows.

between the two modalities. The fusion process is expressed as $U(\theta_M) \cdot U(\theta_u)|X_i\rangle$.

6) Step 6: Model Output and Optimization: Following the multimodal feature fusion, specific qubits are measured, and the expectation value of a quantum observable O is computed as $E(O) = \mathrm{Tr}[O \cdot U(\theta_M) \cdot U(\theta_u)|X_i\rangle\langle X_i|U(\theta_u)^\dagger \cdot U(\theta_M)^\dagger]$. The expectation value $E(O)$ is mapped to the predicted output $\hat{y}_i$. The loss function $L(\hat{y}_i, y_i)$ quantifies the difference between the prediction output and the true label y_i, and the parameters θ are optimized iteratively through backpropagation.

B. Multimodal Data Preprocessing

In this section, the multimodal data preprocessing block is delineated in two stages: first, images and texts are transformed into vector representations, and second, these vectors are encoded into quantum states.

1) Vectorization of Multimodal Data:

a) Vectorization of Image Data: Multimodal data, including visual imagery and natural language, must be converted into vector formats compatible with classical computing systems. To prepare for image data vectorization, the raw data is typically processed through a series of preprocessing steps: resizing to standard dimensions, center cropping to reduce extraneous backgrounds, and normalization to standardize pixel values. After preprocessing, two primary vectorization methods are prevalent: traditional-based and deep learning-based. In the traditional-based methods, such as principal component analysis [52] and histogram of oriented gradients [53], the focus is on manually engineered features, which may falter in capturing the complex, high-level patterns and be sensitive to variations in image conditions. As alternative methods, deep learning-based approaches utilize neural networks or autoencoders to autonomously learn features, thus achieving more sophisticated and abstract representations of image data. Specifically, given a preprocessed image $I_i \in \mathbb{R}^{w \times h \times c}$, where w, h, and c denote the width, height, and number of color channels, respectively, it is transformed into a corresponding image vector $X_i^I \in \mathbb{R}^m$ by a deep learning-based method $g_\theta(\cdot)$

$$X_i^I = g_\theta(I_i) = \left[x_{i,0}^I, x_{i,1}^I, \ldots, x_{i,m-1}^I\right]^T \tag{5}$$

where m is the dimensionality of the feature space after transformation.

b) Vectorization of Text Data: Textual data, due to its semantic intricacies, requires thorough preprocessing. Initially, this involves cleaning the raw text data, including punctuation removal, case normalization, tokenization, stop word elimination, and lemmatization. Subsequently, these cleaned texts are transformed into vector representations by two dominant methods: discrete and distributed representations. In discrete representation methods, such as the typical one-hot encoding, a vector of length comparable to the vocabulary size is assigned to each word, where the element at the word index is set to one, and all others are set to zero. While this method is simple, it becomes increasingly resource-intensive as the vocabulary size grows.

As another fundamental text representation method, distributed word representation efficiently encodes textual data into a dense and low-dimensional vector space [54]. Specifically, a vocabulary V is initially established from the text data T, where each word is assigned a unique index. An embedding matrix $E \in \mathbb{R}^{|V| \times d}$ is subsequently defined, with $|V|$ representing the vocabulary size and d denoting the embedding space dimensionality. For a word w in V, its corresponding vector representation e_w is extracted as $e_w = E_{\text{index}(w)}$. This matrix E is then integrated into neural network architectures and is optimized during the training phase. Finally, the embedding X_i^T for a sentence T_i is derived by applying a fusion function $f(\cdot)$, which aggregates the vector representations of each word in the sentence into a single unified vector

$$X_i^T = f(T_i) = \left[x_{i,0}^T, x_{i,1}^T, \ldots, x_{i,p-1}^T\right]^T \tag{6}$$

where p indicates the resultant dimensionality of the sentence feature space post-fusion.

2) Quantum Encoding: To process classical data with quantum computing systems, it is imperative to encode it into its

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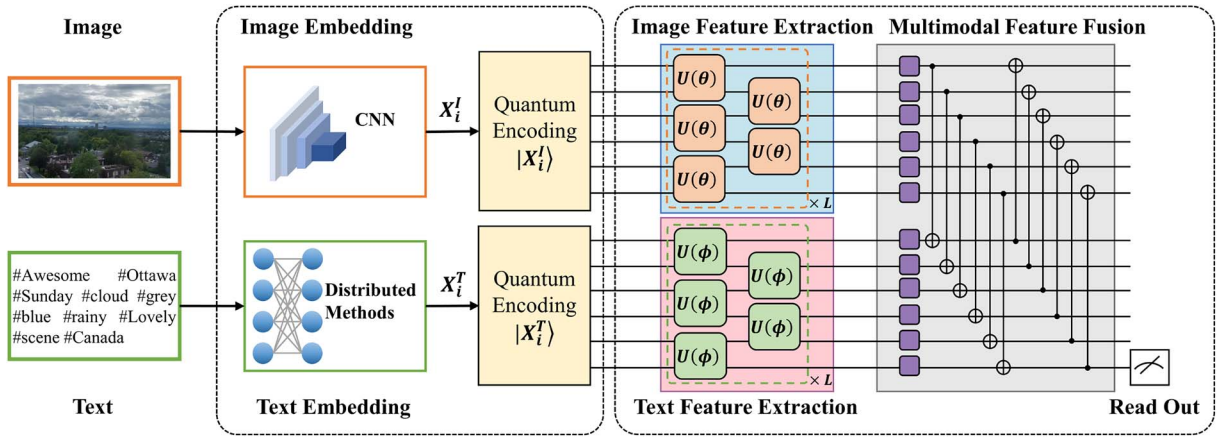


Fig. 4. Comprehensive design of the QMNN framework. The framework integrates image and text preprocessing and embedding, quantum encoding, single-modal feature extraction, multimodal feature fusion, and model readout for sentiment classification.

quantum representation $|X\rangle$. Basis and amplitude encoding are predominantly utilized among the quantum encoding strategies. In the basis encoding method, classical binary data is encoded directly into quantum states utilizing $|0\rangle$ and $|1\rangle$. While this method is straightforward, it becomes qubit-intensive for large or high-precision datasets, limiting its scalability.

As another important quantum encoding method, the amplitude encoding approach encodes classical data into the probability amplitudes of a quantum state. Specifically, a classical vector $x = [x_0, \ldots, x_{2^n-1}]^T$ in a 2^n dimensional space is encoded into a quantum state $|x\rangle$ as follows:

$$|x\rangle = \frac{1}{C}\sum_{i=0}^{2^n-1} x_i\,|i_b\rangle \tag{7}$$

where $C = \sqrt{\sum_{i=0}^{2^n-1}|x_i|^2}$ is the normalization constant, and $|i_b\rangle$ denotes the quantum state corresponding to the binary representation of the integer i, with i spanning the range from 0 to $2^n - 1$. For the vectors X_i^I and X_i^T, derived from image and text data, respectively, the quantum encoding $|X_i^I\rangle$ and $|X_i^T\rangle$ can be formalized as

$$\begin{aligned}\left|X_i^I\right\rangle &= \frac{1}{C_I}\left(x_{i,0}^I|0\ldots0\rangle + \cdots + x_{i,m-1}^I|1\ldots1\rangle\right)\\ \left|X_i^T\right\rangle &= \frac{1}{C_T}\left(x_{i,0}^T|0\ldots0\rangle + \cdots + x_{i,p-1}^T|1\ldots1\rangle\right)\end{aligned} \tag{8}$$

where $C_I = \sqrt{\sum_{j=0}^{m-1}\left|x_{i,j}^I\right|^2}$ and $C_T = \sqrt{\sum_{k=0}^{p-1}\left|x_{i,k}^T\right|^2}$ are the normalization constants for the image and text quantum states, respectively. Notably, a quantum system with n qubits can store up to 2^n complex-numbered probability amplitudes, signifying that the system's storage capacity scales exponentially with the number of qubits.

In summary, in this article, image data vectors are derived through a deep learning-based approach, while text data vectors are generated via a distributed representation method. These vectors are then encoded into quantum states utilizing the amplitude encoding technique, completing the preprocessing phase required for subsequent quantum circuit applications.

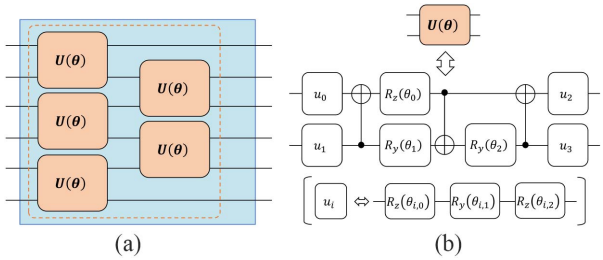


Fig. 5. Unimodal feature extraction block. (a) a quantum convolutional layer architecture. (b) a universal 2-qubit gate.

C. Quantum Multimodal Neural Network Framework

This subsection elucidates the QMNN framework that encompasses the unimodal feature extraction blocks, the multimodal feature fusion block, and the readout, as depicted in Fig. 4. The design of this framework aims to achieve multimodal sentiment analysis by employing PQCs.

1) Unimodal Feature Extraction Block: Unimodal feature extraction blocks are designed to learn features from image-text pairs: an image feature extraction block for visual features and a text feature extraction block for textual features. Inspired by the established CNN-Multi [45] and DNN-LR [55] models, we utilize quantum convolutional layer architecture to learn respective visual and textual features. Classical CNNs are effective in image and text classification because their convolutional layers perform local connections, where the convolution kernels interact with parts of the data to extract local features, and they demonstrate weight sharing by using the same convolution kernel parameters across the entire layer. The quantum convolutional layer architecture employs a PQC to mirror classical convolution effects, as shown in Fig. 5(a). These layers consist of 2-qubit quantum gates, U, that affect adjacent qubits to establish local connections, analogous to classical convolution kernels. The weight-sharing mechanism is implemented by maintaining uniform parameters across all U gates within the layer.

Particularly, the 2-qubit quantum gate U, at the core of the quantum convolutional layer, is universal and capable of

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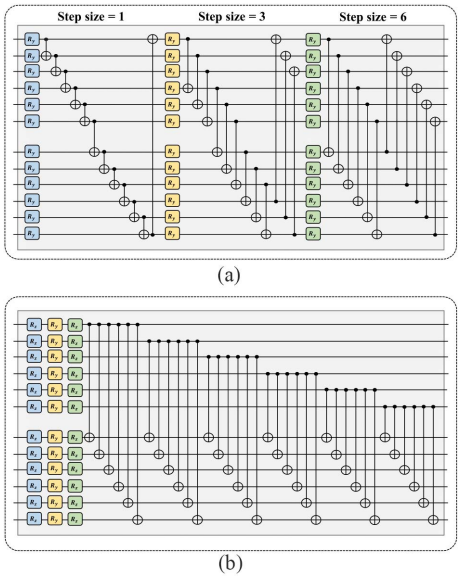


Fig. 6. Multimodal feature fusion blocks. (a) circuit-block as fusion block. (b) all-to-all as fusion block.

executing arbitrary quantum state transformations, enhancing the expressiveness of the PQCs. Given that currently available quantum computing platforms only provide elementary quantum elements, it is essential to decompose this universal gate $U \in SU(4)$ into a series of basic quantum gates as outlined in Table I. In this article, the decomposition strategies from [56] and [57], aimed at minimizing the number of basic gates, are adopted. Specifically, the decomposition of $U \in SU(4)$ consists of three single-qubit rotation gates, three CNOT gates, and four universal single-qubit gates $u_i \in SU(2)$, as depicted in Fig. 5(b). Furthermore, each u_i gate is decomposed into three single-qubit rotation gates. This approach to decomposing universal 2-qubit gates significantly enhances the compactness and practical implementation of quantum convolutional layers.

2) Multimodal Feature Fusion Block: Multimodal feature fusion blocks are designed to integrate textual and visual representations derived from unimodal feature extraction blocks. In classical methods, fully connected layers are commonly used to facilitate this integration, effectively combining features from different modalities. Inspired by this classical framework, we introduce a quantum multimodal feature fusion block that utilizes a quantum fully connected layer architecture. Specifically, this block is tailored to fuse features extracted from unimodal feature extraction blocks, with an emphasis on exploring the interactions between visual and textual representations to generate the final fusion representation.

Quantum fully connected layer architecture can be implemented using a variety of configurations [58], [59]. This article primarily focuses on the most popular and widely recognized configurations [59]: "circuit-block" and "all-to-all," as illustrated in Fig. 6. It should be noted that our design diverges somewhat from [59] to suit our QMNN model better. Specifically, multimodal feature fusion blocks based on PQC are designed to integrate textual and visual representations efficiently,

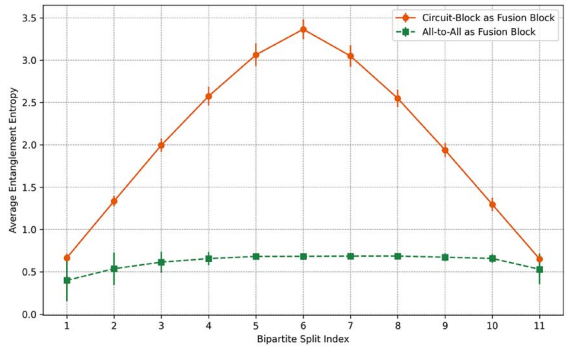


Fig. 7. Average entanglement entropies w.r.t. different fusion blocks.

which requires strong entanglement capability within the block. Therefore, we employ entanglement entropy as a metric to evaluate the entanglement level of the quantum fully connected layer architecture [58]. This evaluation aids in determining the most suitable architecture for the multimodal feature fusion block within our QMNN model.

Specifically, the "von Neumann entanglement entropy" is employed as a measure to evaluate the entanglement capability [60]. Consider a quantum system of n qubits represented by the state $\rho = |\psi\rangle\langle\psi|$. For a bipartite division of this system into subsystems A and B, with n_A and $n_B = n - n_A$ qubits, respectively, the entanglement entropy for subsystem A is determined using its reduced density matrix $\rho_A = \mathrm{Tr}_B[\rho]$. The entanglement entropy, $S(\rho_A)$, is given by the expression

$$S(\rho_A) = -\mathrm{Tr}[\rho_A \log \rho_A] \tag{9}$$

where the logarithm is taken to the natural base e. This measure effectively quantifies the extent of entanglement between subsystem A and its complement B. A value of $S(\rho_A) = 0$ indicates a product state with no entanglement, whereas maximal entanglement between the subsystems results in $S(\rho_A) = n_A \log(2)$.

Based on (9), we analyze the entanglement entropy of quantum fully connected layer architectures with the "circuit-block" and "all-to-all" configurations. The input states are initialized to $|0\rangle^{\otimes n}$, with the parameters of PQCs being randomly sampled from a uniform distribution over $[0, 2\pi]$. The average entanglement entropy is calculated over 100 trials. In addition, we systematically explore the entanglement distribution by partitioning the system into all possible bipartite splits of the n qubits, where n is 12. The results of this comprehensive analysis are depicted in Fig. 7. From Fig. 7, one can observe that the entanglement capability of the "circuit-block" configuration exceeds that of the "all-to-all" configuration. Notably, the highest entanglement entropy is observed in the central bipartite splitting. Therefore, the quantum fully connected layer with the "circuit-block" configuration is used as the multimodal feature fusion block in the QMNN model.

3) Readout: The objective of multimodal sentiment classification is to predict the sentiment label $y \in \{\text{Positive}: 1, \text{Negative}: 0\}$ for a given image-text pair $\{I, T\}$. After performing the quantum circuit of the multimodal feature fusion

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block, the classical information is extracted from the output qubit by applying the projective measurement method. Specifically, by repeatedly executing the entire circuit and using the quantum observable O to measure the system, the measurement outcome can be expressed as the expectation value of O

$$\mathbb{E}(O) = \mathrm{Tr}\left[O \cdot U(\theta) \cdot \rho_{in} \cdot U(\theta)^{\dagger}\right] \tag{10}$$

where $\rho_{\text{in}} = |X_{\text{in}}\rangle\langle X_{\text{in}}|$ represents the input quantum state of the image-text pair. $U(\theta)$ symbolizes the entire unitary operation of the QMNN model, with θ indicating the parameter vector, each element of which corresponds to a quantum gate ($R_x(\theta)$, $R_y(\theta)$, or $R_z(\theta)$). The quantum observable O is chosen as $O = I^{\otimes(n-1)} \otimes Z$ in this article. This configuration ensures that the expectation value $\mathbb{E}(O)$ falls within the range $[-1, 1]$, allowing it to serve as the objective function for the QMNN model. The expectation value is subsequently processed through a sigmoid function, σ, to map it into the probabilistic range $[0, 1]$. The final predicted sentiment label, $\hat{y}$, is determined by comparing the sigmoid-processed expectation value to a threshold of $(1/2)$

$$\hat{y} = \begin{cases} 1, & \text{if } \sigma(\mathbb{E}(O)) \geq \frac{1}{2} \\ 0, & \text{if } \sigma(\mathbb{E}(O)) < \frac{1}{2}. \end{cases} \tag{11}$$

D. Optimization

1) Notation: Before moving on, the notation used in this subsection is defined as follows: $D^{\text{train}} = \left\{\left(I_i^{\text{train}}, T_i^{\text{train}}, y_i^{\text{train}}\right)\right\}_{i=0}^{S-1}$ and $D^{\text{test}} = \left\{\left(I_j^{\text{test}}, T_j^{\text{test}}, y_j^{\text{test}}\right)\right\}_{j=0}^{Q-1}$ represent the training and test sets, respectively, where $I_i^{\text{train}}, I_j^{\text{test}} \in \mathbb{R}^m$, $T_i^{\text{train}}, T_j^{\text{test}} \in \mathbb{R}^p$, $y_i^{\text{train}}, y_j^{\text{test}} \in \{0, 1\}$, S and Q indicates the respective set sizes, m and p indicate the dimensionality of the image and sentence feature space postfusion, respectively. Moreover, the loss function is defined as the binary cross-entropy loss

$$\ell(\theta) = -\frac{1}{S}\sum_{i=0}^{S-1}\left[y_i^{\text{train}} \log(\sigma(\mathbb{E}(O_i))) + (1 - y_i^{\text{train}}) \log(1 - \sigma(\mathbb{E}(O_i)))\right] \tag{12}$$

where $\sigma(\mathbb{E}(O_i))$ is the sigmoid function that processes the expectation value of the observable O for the ith sample in D^{train}.

2) Optimization Algorithm: Similar to the gradient-based optimization methods employed in classical neural networks, the quantum stochastic gradient descent (QSGD) algorithm is introduced to optimize the QMNN model. This algorithm enables the backpropagation mechanism, a crucial process in training neural networks. The specific details of the QSGD algorithm are presented in Algorithm 1.

It is pointed out that the useful parameter shifting rule in [9], [61] is employed in the algorithm to compute the partial derivative of $\mathbb{E}(O)$ to the parameter θ_i

$$\frac{\partial \mathbb{E}(O)}{\partial \theta_i} = \frac{1}{2}\mathrm{Tr}\left[O \cdot U\left(\theta + \frac{\pi}{2}e_i\right) \cdot \rho_{in} \cdot U\left(\theta + \frac{\pi}{2}e_i\right)^{\dagger}\right] - \frac{1}{2}\mathrm{Tr}\left[O \cdot U\left(\theta - \frac{\pi}{2}e_i\right) \cdot \rho_{in} \cdot U\left(\theta - \frac{\pi}{2}e_i\right)^{\dagger}\right] \tag{13}$$

Algorithm 1: QSGD Algorithm

Input: The input $\{\rho_i^{\text{train}} = |X_i^{\text{train}}\rangle\langle X_i^{\text{train}}|\}_{i=0}^{S-1}$;
The parameterized quantum circuit $U(\theta)$;
The quantum observable O, the maximum iteration time T_{max}, and the threshold δ;
The predetermined learning rate η.
Output: The optimized parameter vector θ^*.

```
t ← 0;
Initialize each parameter θ^(t) randomly within [0, 2π];
while t ≤ T_max and ||ℓ(θ^(t))|| ≥ δ do
    Selecting a sample: {ρ_i^train, y_i^train}_{i=0}^{S-1} randomly;
    Computing the gradient:
    ∇ℓ(θ^(t)) = (σ(E(O_i)) − y_i^train) · ∂E(O_i)/∂θ^(t) using the
     chain rule and the parameter shifting rule in (13).
    Updating parameter:
    θ^(t+1) = θ^(t) − η · ∇ℓ(θ^(t));
    t ← t + 1;
end
```

where e_i is a zero vector with its ith element replaced by one.

IV. Simulation

In this section, the proposed QMNN model is evaluated for its application in multimodal sentiment analysis tasks. First, the datasets used and the preprocessing methods employed are detailed. Subsequently, the baseline models and simulation settings are outlined. Finally, simulation results and discussions are provided to highlight the advantages of the proposed model.

A. Dataset and Preprocessing

1) Dataset: The QMNN model is evaluated on two public benchmark datasets, MVSA-Single and MVSA-Multiple, which are widely recognized and commonly used in the multimodal sentiment analysis field [25], [62], [63].

a) MVSA-Single: This dataset consists of 5129 text-image pairs sourced from Twitter. Each pair is labeled by one annotator, who assigns positive, negative, or neutral sentiment polarity labels to the text and the accompanying image, respectively.

b) MVSA-Multiple: This dataset, larger in scale than MVSA-Single, includes 19600 text-image pairs. In addition, each pair in MVSA-Multiple is independently labeled by three annotators, with each annotator assigning separate sentiment polarity labels to the text and the image.

2) Preprocessing: For a fair comparison, the preprocessing methods for both datasets are adapted from [25], [63], [64]. Specifically, for the MVSA-Multiple dataset, the following preprocessing steps are implemented to ensure the reliability of annotations and minimize subjective bias.

a) Step 1: Determining Single-Modality Polarity: The definitive label for each modality is established based on a majority vote among the three annotators. An image or text is considered valid only if at least two of the three annotators agree on the same label.

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TABLE II
STATISTICS OF THE PROCESSED MVSA DATASETS

Dataset	Positive	Negative	Total
MVSA-Single	2683	1358	4041
MVSA-Multiple	11318	1298	12616

b) Step 2: Removing Polarity-Inconsistent Text-Image Pairs: Text-image pairs are removed if they exhibit direct polarity conflicts, where one label is positive and the other negative. In the case where one label is neutral and the other is either positive or negative, the overall sentiment polarity of the pair is regarded as positive or negative.

c) Step 3: Retaining Binary Polarity Pairs: Text-image pairs with positive or negative sentiment polarity labels are retained.

For the MVSA-Single dataset, the preprocessing follows steps 2 and 3. The preprocessed MVSA-Single and MVSA-Multiple datasets are thus prepared for our simulation, as illustrated in Table II.

B. Baseline Models and Simulation Settings

1) Baseline Models: We compare our proposed QMNN model with several state-of-the-art models, including CNN-MKL (CNNs with multiple kernel learning) [44], DNN-LR (deep CNNs with logistic regression) [55], CNN-Multi (CNNs for multimedia) [45], CNN-Multi-PANet (CNN-Multi-Path Aggregation Network), RMNN (residual merged neural network) [64], all of which are classical versions most directly relevant to our model.

a) CNN-MKL: This model leverages deep CNNs to extract visual and textual features. These features are then fused by a multiple kernel learning (MKL) module that employs various kernel functions to handle the heterogeneous multimodal data effectively.

b) DNN-LR: This model utilizes deep convolutional neural networks (DNNs) to extract visual and textual features independently. Subsequently, logistic regression is employed to fuse these multimodal features, thus enhancing sentiment classification in diverse contexts.

c) CNN-Multi: This model adopts two individual CNN architectures for the separate extraction of visual and textual features. These extracted features are combined and processed through an additional CNN with four fully connected layers, aiming to examine the internal relationships between textual and visual modalities for effective sentiment analysis.

d) CNN-Multi-PANet: CNN-Multi-PANet integrates the feature pyramid architecture from PANet [65] into the CNN-Multi model. Additional convolutional layers are introduced in the unimodal feature extraction blocks, followed by interpolation, element-wise max pooling, and adaptive max pooling to align and fuse features across multiple levels. This model leverages the multiscale feature extraction capability of PANet to provide a more comprehensive representation of multimodal data.

e) RMNN: This model employs CNNs to extract visual and textual features separately. These features are integrated using a residual model along with two distinct merging approaches, Early-RMNN and Late-RMNN. This strategy is instrumental in uncovering deeper and more discriminative features, improving the performance of multimodal sentiment analysis.

It is worth noting that so far, few quantum versions of neural networks based on PQCs have been proposed for processing multimodal data. In light of this, the proposed QMNN is primarily benchmarked against the most directly relevant classical version in simulations to establish a fundamental comparison. Moreover, to thoroughly evaluate the capabilities of the proposed QMNN, a series of in-depth evaluations is conducted. These evaluations focus on exploring multimodal fusion methodologies, conducting ablation studies, and analyzing quantum noise using various configurations of the proposed QMNN, as detailed below.

2) Simulation Settings: In this section, the simulation setup is outlined, including the model configurations and parameter settings.

a) Model Configurations: Currently, the available quantum computing platforms are constrained by providing only a limited number of qubits. Moreover, simulating quantum functions on classical computers leads to an exponentially growing state space with an increasing number of qubits, creating significant computational challenges beyond 20 qubits, and making it difficult to support large-scale and high-dimensional data processing tasks. Indeed, this problem can be well addressed when more powerful real quantum processors are built. Therefore, this article simulates the entire quantum computation procedure on a classical computer and focuses on low-dimensional multimodal datasets for bipolar sentiment analysis tasks, as discussed in Section IV-A.

Specifically, in the MVSA-Single and MVSA-Multiple datasets, both text and images are preprocessed and represented in a 64-dimensional feature space, as detailed in Section III-B. The visual features are extracted using a convolutional autoencoder, while the textual features are obtained through an embedding layer. Following this, 12 qubits and 66 parameters are required for the QMNN architecture, as illustrated in Fig. 4. The detailed configurations of unimodal feature extraction blocks and the multimodal feature fusion block have been discussed in Section III-C. In addition, the baseline models implement the configurations specified in their respective references. To ensure a rigorous and fair comparison, all models employ consistent data preprocessing methods for texts and images, as given in Section III-B, utilize identical training and testing protocols, and follow uniform evaluation metrics. For detailed configurations of each model, please refer to Table III.

b) Parameter Settings: In the experiments, the dataset is divided into training and test sets using stratified fivefold cross-validation to ensure balanced class distribution in each fold. All experiments are performed under the same random seed to guarantee reproducibility and consistency. Each quantum network layer is configured as a single-layer structure. The quantum circuit parameters are uniformly initialized within the

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TABLE III
MODEL CONFIGURATIONS

Models	Configurations			
	Text Feature Extraction	Image Feature Extraction	Fusion Feature Extraction	Others
CNN-MKL	① GloVe-200d embeddings are applied to the nn.Embedding layer. ② 2 convolution layers (kernels: 3×3, 2×2); 2 max-pooling layers (stride: 2); 2 fully connected layers (256d output).	The VGG16 pre-trained model is used up to the 24th layer; 1 fully connected layer (256d output).	① Concatenate the obtained text and image features. ② MKL module has 8 kernels (5 RBF, gammas: 0.01-0.05; 3 Polynomial, degrees: 2-4); 1 output layer (1d output).	Dropout: 0.5; Activation: ReLU (CNN); Sigmoid (output layer).
CNN-Multi	① GloVe-200d embeddings. ② 3 convolution layers (kernels: 5×5, 4×4, 3×3); 3 max-pooling layers (stride: 2); 2 fully connected layers (256d output).	5 convolution layers (kernels: 17×17, 13×13, 7×7, 5×5, 3×3); 5 max-pooling layers (stride: 2); 3 fully connected layers (256d output).	① Concatenate the obtained text and image features (512d output). ② 4 fully connected layers (256d output); 1 output layer (1d output).	Dropout: 0.5; Activation: ReLU (CNN); Sigmoid (output layer).
CNN-Multi-PANet	① GloVe-200d embeddings. ② 3 convolution layers (kernels: 5×5, 4×4, 3×3); 3 max-pooling layers (stride: 2); ③ 2 additional convolution layers (kernel: 1×1) applied after the 3rd and 2nd convolution layers in ②, respectively; Align feature maps from the 2 additional convolution layers; Element-wise max-pooling; 1 adaptive max-pooling layer; 1 fully connected layer (256d output).	① 5 convolution layers (kernels: 17×17, 13×13, 7×7, 5×5, 3×3); 5 max-pooling layers (stride: 2). ② 2 additional convolution layers (kernel: 1×1) applied after the 5th and 4th convolution layers in ①, respectively; Align feature maps from the 2 additional convolution layers; Element-wise max-pooling; 1 adaptive max-pooling layer; 1 fully connected layer (256d output).	① Concatenate the obtained text and image features (512d output). ② 4 fully connected layers (256d output); 1 output layer (1d output).	Dropout: 0.5; Activation: ReLU (CNN); Sigmoid (output layer).
DNN-LR	① GloVe-200d embeddings. ② 3 convolution layers (kernels: 3×3, 4×4, 5×5); 3 max-pooling layers (stride: 2); 1 fully connected layer (128d output).	2 convolution layers (kernel: 3×3); 2 batch normalization layers; 2 max-pooling layers (stride: 2); 2 locally connected layers; 1 drop-connected layer (drop = 0.5); 1 fully connected layer (128d output).	① Concatenate the obtained text and image features (256d output). ② 1 fully connected layer (256d output); 1 output layer (1d output).	Dropout: 0.5; Activation: ReLU (CNN); Sigmoid (output layer).
Early-RMNN	① GloVe-200d embeddings. ② 1 convolution layer (kernel: 5×5); 1 batch normalization layer; 1 max-pooling layer (stride: 3); 1 fully connected layer (output: text_shallow with 256d). ③ 1 fully connected layer (input: text_shallow, output: text_deep with 256d). ④ 1 fully connected layer (input: text_shallow + text_deep, output: text_res with 256d).	① The VGG16 pre-trained model is used up to the 24th layer; 1 batch normalization layer; 1 fully connected layer (output: image_shallow with 256d). ② 1 fully connected layer (input: image_shallow, output: image_deep with 256d). ③ 1 fully connected layer (input: image_shallow + image_deep, output: image_res with 256d).	① Concatenate the text_res and image_res (512d output). ② 1 fully connected layer (256d output); 1 output layer (1d output).	Dropout: 0.5; Activation: ReLU (CNN); Sigmoid (output layer).
Late-RMNN	① GloVe-200d embeddings. ② 1 convolution layer (kernel: 5×5); 1 batch normalization layer; 1 max-pooling layer (stride: 3); 1 fully connected layer (output: text_shallow with 256d). ③ 1 fully connected layer (input: text_shallow, output: text_deep with 256d).	① The VGG16 pre-trained model is used up to the 24th layer; 1 batch normalization layer; 1 fully connected layer (output: image_shallow with 256d). ② 1 fully connected layer (input: image_shallow, output: image_deep with 256d).	① Concatenate the text_shallow and image_shallow (512d); 1 fully connected layer (output: combined_shollow with 256d). ② Concatenate the text_deep and image_deep (512d output); 1 fully connected layer (output: 256d combined_deep). ③ 1 fully connected layer (input: combined_shallow + combined_deep, output: 256d); 1 output layer (1d output).	Dropout: 0.5; Activation: ReLU (CNN); Sigmoid (output layer).
QMNN	① Text data are transformed into 64-dimensional vectors by an nn.Embedding layer. ② Text features are extracted using a unimodal feature extraction block comprised of 5 universal 2-qubit gates $U(\theta)$, sharing 15 parameters. Details of $U(\theta)$ are provided in Fig. 5.	① Image data are reduced to 64-dimensional vectors using a Convolutional Autoencoder that consists of the following components: 2 convolution layers (kernel: 3×3); 2 max-pooling layers (stride: 2). ② Similar to text, a unimodal feature extraction block with 5 universal 2-qubit gates $U(\theta)$ is used to extract image features.	A multimodal feature fusion block with a circuit-block configuration is employed to merge text and image features. This block includes 36 individual R_y gates, applied sequentially to each qubit followed by CNOT gates arranged in strides of 1, 3, and 6. Configuration details are in Fig. 6(a).	Activation: ReLU (CNN); Others are detailed in Sections III-C and III-D.

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$[0, 2\pi)$ range. During training, optimization is conducted over 50 epochs using a stochastic gradient descent algorithm with a batch size of 16 and a learning rate of 0.01. An early stopping mechanism is implemented to prevent overfitting. Quantum circuit simulations are carried out using the PennyLane platform, which is widely recognized and utilized for PQC simulations and QML implementations [20], [23], [60], due to its support for circuit optimization and seamless integration with machine learning frameworks, making it highly suitable for our experimental setting.

To evaluate the performance of the proposed model, two primary evaluation metrics, i.e., test accuracy and weighted F1 score, are implemented and are defined as follows:

$$\text{Test Accuracy} = \frac{1}{Q}\sum_{i=0}^{Q-1}[\hat{y}_i = y_i] \tag{14}$$

where $\hat{y}_i$ is the model's predicted outcome as calculated by the formula (11), y_i denotes the true label, and Q represents the size of the test set. The square brackets $[\cdot]$ act as an indicator function, assigning a value of 1 when $\hat{y}_i$ matches y_i, and 0 otherwise.

$$\text{Weighted F1 Score} = \sum_{k=1}^{K} w_k \cdot \frac{2 \cdot \text{precision}_k \cdot \text{recall}_k}{\text{precision}_k + \text{recall}_k} \tag{15}$$

where K is the number of classes, w_k is the fraction of the dataset consisting of instances from class k, and precision and recall for each class k are defined as

$$\text{precision}_k = \frac{TP_k}{TP_k + FP_k}$$
$$\text{recall}_k = \frac{TP_k}{TP_k + FN_k}.$$

Here, FN_k, FP_k, and TP_k denote the number of false negatives, false positives, and true positives for class k, respectively.

Both metrics are computed at the end of each epoch for each fold. The best test accuracy and its corresponding weighted F1 score across epochs are noted for the fold. After completing the five-fold cross-validation, the means and standard deviations of these best test accuracies and their corresponding weighted F1 scores are recorded across all folds.

C. Simulation Results and Discussion

In this section, we present the simulation results of the QMNN model, comparing it with state-of-the-art classical models on benchmark datasets. In addition, we analyze the internal performance in terms of entanglement and integrity, while the external performance focuses on robustness and scalability. Specifically, entanglement is examined through the impact of different configurations of multimodal feature fusion blocks. Integrity is evaluated via a series of ablation studies that reveal the contribution of each component. Robustness is assessed by analyzing the model's tolerance to both quantum noise and incomplete input data. Scalability is explored by examining the model's ability to extend in both width and depth to accommodate more complex scenarios.

1) Comparison with Baseline Methods: The proposed QMNN model is benchmarked against the most directly relevant and state-of-the-art classical baseline models, establishing a fundamental comparison for multimodal sentiment analysis. While these baseline models are closely related to our QMNN, it is important to note two significant distinctions.

a) Significant Variation in Input Data Dimensionality: Classical models benefit from high-dimensional input data, which allows them to learn rich textual and visual features and interrelations, leading to more effective sentiment classification. In contrast, constrained by the limited number of qubits available on current quantum computing platforms, the QMNN processes multimodal data in lower dimensions. In our comparative experiments, classical models processed image and text data at a minimum dimensionality of 128, whereas the QMNN handled these data at only 64 dimensions each.

b) Considerable Differences in Parameter Count: In QMNN, quantum convolutional layers within unimodal feature extraction blocks utilize universal 2-qubit gates that share parameters across each layer. This parameter-sharing strategy substantially reduces the overall parameter count of the model. Specifically, a configuration with l_u quantum convolutional layers results in a parameter count of $15 \times l_u$ for each unimodal feature extraction block. Additionally, l_m quantum fully connected layers in the multimodal feature fusion block require $3 \times n \times l_m$ parameters, where n represents the number of qubits in the QMNN. Furthermore, the total parameter count for the QMNN amounts to $2 \times 15 \times l_u + 3 \times n \times l_m$, which is far lower compared to classical baseline models. In our experiments, the proposed QMNN model employs a single layer configuration for all quantum layers, resulting in a minimal parameter count of only 66.

The experimental results, which compare the proposed QMNN with state-of-the-art baseline models on the MVSA-Single and MVSA-Multiple datasets, are listed in Table IV.

From Table IV, one can observe that the QMNN model surpasses all baseline models in terms of test accuracy and exhibits competitive performance in weighted F1 score on both the MVSA-Single and MVSA-Multiple datasets, highlighting its effectiveness in processing multimodal sentiment analysis for large scale datasets. This superior performance is mainly attributed to two aspects of QMNN. First, the quantum convolutional layer architecture is utilized within both the image and text feature extraction blocks, facilitating the efficient extraction of visual and textual features. Second, the quantum fully connected layer with strong entanglement capabilities is employed in the multimodal feature fusion block, enabling efficient integration of textual and visual features, which enhances sentiment classification. In addition, one can also observe that QMNN shows a more pronounced advantage on the MVSA-Single dataset than the MVSA-Multiple dataset. This difference is attributed to the more balanced class distribution of tweets in MVSA-Single, despite the larger dataset size in MVSA-Multiple.

In summary, the QMNN model is adept at learning textual and visual features and their interactions, utilizing low-dimensional multimodal data and fewer parameters, which

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TABLE IV
COMPARATIVE RESULTS WITH BASELINE MODELS*

Method	Parameters	MVSA-Single		MVSA-Multiple	
		Test Accuracy (%)	Weighted F1 Score (%)	Test Accuracy (%)	Weighted F1 Score (%)
CNN-MKL	27.56M	69.78 ± 1.78	66.44 ± 2.48	88.63 ± 0.30	85.79 ± 0.32
CNN-Multi	5.30M	72.11 ± 3.10	68.56 ± 7.95	90.00 ± 0.28	**86.95 ± 1.78**
CNN-Multi-PANet	8.97M	72.85 ± 8.00	64.74 ± 14.40	89.71 ± 0.02	84.85 ± 0.03
DNN-LR	51.82M	71.66 ± 4.31	65.51 ± 10.18	89.85 ± 0.26	85.71 ± 1.72
Early-RMNN	27.69M	72.28 ± 0.97	70.75 ± 1.16	89.93 ± 0.28	85.73 ± 1.09
Late-RMNN	27.75M	72.68 ± 0.66	70.48 ± 0.57	89.95 ± 0.34	85.53 ± 1.02
QMNN	66	**73.27 ± 0.50**	**70.88 ± 1.03**	**90.08 ± 0.45**	85.96 ± 1.34

Note: *The comparative experiments are conducted under noise-free conditions to ensure a fair comparison with classical models. The bold entries indicate the best results for each metric under the respective dataset.

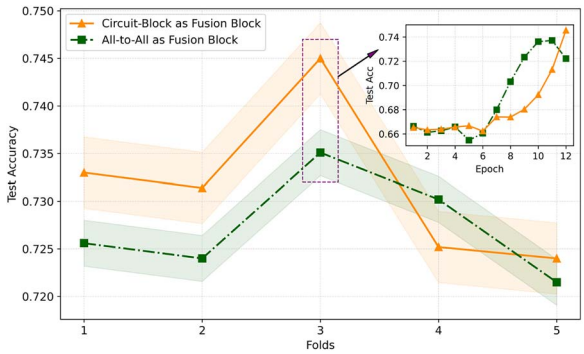


Fig. 8. Test accuracies w.r.t. different fusion blocks. Figure illustrates the results across five-fold cross-validation, with an inset detailing the performance changes over the first 12 epochs of fold 3.

results in exceptional performance across all benchmarks for multimodal sentiment classification.

2) Entanglement (Internal Performance): Given the crucial role of the multimodal fusion block in multimodal sentiment analysis, we investigate the impact of its configuration on the entanglement performance of the QMNN model. Inspired by classical models that commonly employ fully connected layers in their fusion blocks, we explore quantum fully connected layers with two distinct configurations in the multimodal feature fusion block. Specifically, these two configurations, derived from the "circuit-block" and "all-to-all" structures detailed in [58], are appropriately modified for effective integration into the QMNN framework. In the "circuit-block," R_y gates are applied to all qubits, followed by CNOT gates with control steps of 1, 3, and 6. In the "all-to-all," universal single-qubit gates are applied to all qubits, with each qubit designated for image feature extraction directly connected to every qubit designated for text feature extraction via CNOT gates. The implementation of these configurations is illustrated in Fig. 6. Apart from these different configurations in the multimodal feature fusion block, all other experimental settings remain consistent with those described in Section IV-B. The results of these experiments on the MVSA-Single dataset are depicted in Fig. 8.

From Fig. 8, one can observe that the QMNN model, utilizing the "circuit-block" as the fusion block, consistently achieves higher test accuracies across most folds compared to the "all-to-all" configuration. In addition, the inset provides a detailed view of the performance changes over the first 12 epochs of fold 3, illustrating that despite initial fluctuations, the "circuit-block" exhibits steady improvement and eventually surpasses the performance of the "all-to-all" configuration. These results align with the entanglement capability analysis of the multimodal feature fusion block discussed in Section III-C, indicating that PQCs with stronger entanglement capabilities are more effective in fusing visual and textual features and learning their interrelations, thereby enhancing the overall performance of the model.

3) Integrity (Internal Performance): To further validate the effectiveness of each component and its contribution to the integrity of the proposed QMNN, we conduct a series of ablation experiments on the MVSA-Single dataset. The QMNN mainly consists of an image feature extraction block, a text feature extraction block, and a multimodal feature fusion block, as depicted in Fig. 4. In these ablation studies, each of these blocks is selectively removed from the QMNN to assess their individual contributions. Specifically, the "QMNN w/o Image" configuration removes the image feature extraction block, processing only the text data from the image-text pairs. Conversely, the "QMNN w/o Text" setup excludes the text feature extraction block, focusing exclusively on the image data within the pairs. Notably, in the "QMNN w/o Strong Entanglement Fusion" variant, the multimodal feature fusion block is replaced by a simple fusion approach employing sequential CNOT gates on adjacent qubits. All other experimental setups are consistent with those described in Section IV-B. The results of these experiments are illustrated in Table V.

From Table V, one can observe that the proposed QMNN model consisting of all blocks achieves the best test accuracy on the MVSA-Single dataset, and removing any block results in a discernible reduction in classification effectiveness. This demonstrates that each block is indispensable and contributes jointly to the final performance. One can also observe that the

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TABLE V
ABLATION ANALYSIS RESULTS*

Method	Test Accuracy (%)	N_q	N_p
QMNN w/o Image	72.85 ± 0.74	6	15
QMNN w/o Text	67.43 ± 0.72	6	15
QMNN w/o SE Fusion	72.95 ± 0.74	12	30
QMNN	**73.27 ± 0.50**	12	66

Note: *N_q and N_p denote the number of qubits and the number of parameters, respectively.

TABLE VI
QUANTUM CHANNELS*

	BitFlip	PhaseFlip
Kraus Matrices	$K_0 = \sqrt{1-p}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$ $K_1 = \sqrt{p}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$	$K_0 = \sqrt{1-p}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$ $K_1 = \sqrt{p}\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}$

Note: *$p \in [0, 1]$ is noise level.

fusion of image and text features in the QMNN model enhances its ability to capture more detailed and nuanced sentiment profiles, surpassing the performance of models that focus on single-modality inputs. In addition, text features emerge as more influential than image features in the MVSA-Single dataset, contributing more noticeably to sentiment classification.

4) Robustness (External Performance): To evaluate the robustness of the QMNN, we conduct a series of experiments to assess the impact of quantum noise and incomplete inputs on the model performance.

a) Impact of Quantum Noise: Quantum noise is modeled using quantum channels, represented by Kraus operators K_i that satisfy $\sum_i K_i^\dagger K_i = I$. For an initial quantum state ρ, the output state $\Phi(\rho)$ following the action of a quantum channel is given by

$$\Phi(\rho) = \sum_i K_i \rho K_i^\dagger. \tag{16}$$

In our experiments, we consider two types of quantum channels: the single-qubit bit flip and phase flip channels. The Kraus matrices for these channels are presented in Table VI.

These quantum channels are applied to each qubit following both the unimodal feature extraction and multimodal feature fusion blocks within the QMNN architecture, with noise levels set at 0%, 1%, 10%, and 20%. It is important to note that the implementation of the quantum channel introduces additional computational overhead. To enhance the operational efficiency, we reduce the vector dimension to 16 for both text and image, which requires a total of 8 qubits to construct the QMNN. The MVSA-Single dataset is divided into an 80–20 training-to-test ratio under the same random seed. All other experimental setups are consistent with those described in Section IV-B. The results of these experiments are illustrated in Fig. 9.

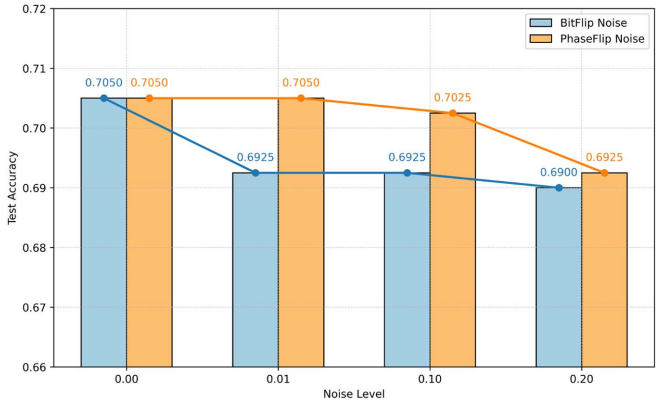


Fig. 9. Test accuracies w.r.t. different quantum noises.

From Fig. 9, one can observe that the QMNN demonstrates significant robustness, effectively adapting to various types of high-level noise. Specifically, for phase flip noise, the classification performance of the QMNN remains stable when the noise level is below 1%. Even with a noise level of 20%, the overall performance degradation is minimal, with only a 1.25% decrease in test accuracy observed. One can also observe that various types of noise affect the model differently at the same noise level. For instance, bit flip noise affects the QMNN marginally more than phase flip noise, while the effect is still slight for the whole model. In summary, the robustness of QMNN ensures reliable performance even in the presence of substantial noise interference.

b) Impact of Incomplete Inputs: Considering real-world scenarios often involve missing or corrupted data, the robustness of the QMNN is further evaluated under incomplete multimodal inputs. Experiments are designed with varying levels of input reduction applied to both text and images to simulate such conditions. In the text data, a fraction of tokens from each sample is randomly removed, with 10% and 25% discarded across different settings. In the image data, random masking is applied, with 20% and 40% of the pixels occluded to simulate partial occlusion. The combined effect is evaluated by simultaneously discarding 10% of the text tokens and 20% of the image pixels, as well as 25% of the text tokens and 40% of the pixels. Apart from these input reduction strategies, all other experimental settings remain consistent with those described in Section IV-B. The results of these experiments on the MVSA-Single dataset are summarized in Table VII.

From Table VII, one can observe that the QMNN demonstrates strong resilience under various levels of input reduction. Masking up to 25% of text tokens or 40% of image pixels leads to only slight decreases in accuracy and F1 scores, indicating the model can retain essential features from incomplete data. When both modalities are reduced simultaneously, the performance remains stable, showcasing the robustness of its multimodal fusion. These results confirm the robustness of QMNN, ensuring reliable outcomes even in scenarios with missing or corrupted inputs.

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TABLE VII
ROBUSTNESS ANALYSIS OF QMNN WITH PARTIAL INPUT MASKING

Masking Scope	Test Accuracy (%)	Weighted F1 Score (%)
Text Mask 10%	71.79 ± 2.58	66.85 ± 6.40
Text Mask 25%	71.74 ± 0.59	68.56 ± 1.53
Image Mask 20%	73.10 ± 0.61	70.39 ± 1.63
Image Mask 40%	71.76 ± 2.30	67.49 ± 6.09
Text 10% + Image 20%	72.26 ± 1.00	69.31 ± 1.51
Text 25% + Image 40%	72.11 ± 0.71	69.20 ± 1.71
No Masking	73.27 ± 0.50	70.88 ± 1.03

TABLE VIII
SCALABILITY ANALYSIS OF QMNN MODEL

Scalability	Configuration	Test Accuracy (%)	Weighted F1 Score (%)
Breadth	8 qubits	72.85 ± 0.63	71.10 ± 1.36
	10 qubits	72.95 ± 1.29	70.73 ± 1.63
	12 qubits	73.27 ± 0.50	70.88 ± 1.03
	14 qubits	73.32 ± 0.47	71.30 ± 0.63
Depth	One layer	73.27 ± 0.50	70.88 ± 1.03
	Three layers	73.47 ± 0.57	71.94 ± 1.18

5) Scalability (External Performance): To comprehensively evaluate the scalability of the QMNN model, we perform a series of experiments to examine how variations in both the breadth and depth of quantum circuits influence model performance. For breadth scalability, QMNN models with different numbers of qubits are configured to assess whether richer visual and textual features can enhance performance. Specifically, vector dimensions of 16, 32, 64, and 128 are considered for both image and text inputs, corresponding to QMNN models requiring 8, 10, 12, and 14 qubits, respectively. For depth scalability, the number of layers in the unimodal feature extraction block of the QMNN model is adjusted to determine whether deeper layers can more effectively capture complex features and improve the model performance. In particular, we compare single-layer and three-layer structures. All scalability evaluations are conducted on the MVSA-Single dataset, with other experimental settings remaining consistent with those outlined in Section IV-B. The detailed results are presented in Table VIII.

From Table VIII, one can observe that both test accuracy and weighted F1 score gradually improve as the number of qubits increases from 8 to 14. This trend reflects that richer visual and textual inputs allow the model to better capture multimodal information. Similarly, the deeper architecture with three layers in the feature extraction block outperforms the one-layer structure, indicating that a more complex network architecture can extract intricate features more effectively. One can also observe that beyond 12 qubits, further increases in the number of qubits yield no significant improvement in performance. Considering the constraints of current quantum platforms, we adopt a configuration of 12 qubits with a single layer as a balanced design in this article, which ensures both computational efficiency and stable performance.

V. CONCLUSION

In this article, we have introduced a novel QMNN model based on PQCs tailored for multimodal sentiment analysis tasks. The model consists of four essential blocks: the multimodal data preprocessing block, which transforms images and texts from multimodal data into vector representations and encodes them into quantum states; the unimodal feature extraction block, which extracts visual and textual features; the multimodal feature block, which efficiently integrates the extracted unimodal features to facilitate feature interaction; and the network optimization block, which fine-tunes the model parameters to enhance performance. Simulations have been performed on public multimodal datasets to demonstrate the superiority of the proposed model. Future research will focus on incorporating additional modalities, such as audio, to enhance interaction among diverse data types. We also plan to explore multitask learning to unlock shared feature representations across tasks and develop optimized architectures to maintain scalability as data complexity grows.

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